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Molecular phylogenetics and microsatellite analysis reveal cryptic species of speckled dace (Cyprinidae: *Rhinichthys osculus*) in Oregon's Great Basin

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ABSTRACT

Speckled dace (*Rhinichthys osculus*) is a small cyprinid that occurs throughout western North America and is the most commonly occurring fish in Oregon. Because of the high genetic and morphological variation in this species across its range, it has been referred to as a species complex; however, no revision to its taxonomy has occurred since 1984. Here, the phylogenetics and population genetics of speckled dace are examined throughout Oregon's Great Basin to describe genetic variation and infer the geographic boundaries between distinct taxonomic entities and populations. We tested the validity of a putative subspecies, Foskett Spring speckled dace, that occurs in a single spring within Warner Valley in Southeast Oregon and is listed Federally as threatened. Dace were collected from Foskett Spring and all surrounding basins containing speckled dace (Warner, Goose Lake, Lake Abert, Silver Lake, and Malheur), as well as Stinking Lake Spring (located within Malheur), created phylogenetic trees from mitochondrial ND2 and nuclear S7 sequence data, and genotyped eight microsatellite loci for population-level analyses. Three highly divergent clades warrant species-level status: Malheur stream dace, Stinking Lake Spring dace, and dace from the other four basins combined. Although Foskett Spring dace were not monophyletic, substantial population structure occurs at the basin-level and separates Foskett Spring dace from other dace in the surrounding Warner Valley. Thus, we recommend ESU status for the isolated population of speckled dace in Foskett Spring. The high, previously unrecognized, taxonomic diversity within this region indicates a need for a range-wide phylogeographic study of speckled dace and an investigation of the morphological distinctiveness of the putative new species.

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1. Introduction

The development of effective conservation and management strategies depends on reliable species designation and identification (Mace, 2004; Wheeler, 2004). However, species boundaries are often poorly understood because closely related species can be difficult to distinguish using the morphological approaches that dominated the twentieth century (Strauss and Bond, 1990; deQueiroz, 2007). Given their extensive polymorphism, plasticity and allometry, fish species can be particularly difficult to distinguish using morphological data (Turner, 1999; Callejas and Ochando, 2001). In regions of complex biogeographic history, with repeated connections, isolations, and invasions, identifying species can be even more difficult. These difficulties can be abetted by a phylogenetic approach (DiBattista et al., 2012; Lopes-Cunha et al., 2012). Genetic and phylogenetic tools allow for increased detection of cryptic diversity (Bickford et al., 2007), which in turn has helped revise the confusing taxonomy of well-known groups of fish (Kon et al., 2007; Griffiths et al., 2010; Craig et al., 2009). Herein, we apply modern molecular tools to help clarify the taxonomic and population boundaries within one of the most perplexing species complexes among North American freshwater fishes.

1.1. Speckled dace taxonomy in Oregon's Great Basin

Speckled dace (*Rhinichthys osculus*) is a species of cyprinid fish that is widespread throughout western North America and may represent a complex of morphologically similar species (Page and Burr, 1998). Populations fitting the current taxonomic concept of *R. osculus* occur as far south as Sonora, Mexico and as far north as British Columbia, Canada (Lee, 1980; Page and Burr, 1998). Populations can be found in many isolated sub-basins and interior

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drainages along the Pacific coast, the Great Basin, and Colorado River basin (Bisson and Bond, 1971 and Bond et al., 1988). This ubiquity results because speckled dace can live in diverse habitats from small springs or streams, to large rivers and deep lakes (Page and Burr, 1998).

Previous phylogeographic studies have revealed high genetic differentiation among the major drainage basins that speckled dace occupy (Oakey et al., 2004; Pfrender et al., 2004; Smith and Dowling, 2008; Billman et al., 2010). Speckled dace also vary geographically in meristics, color, size, and body proportions (Hubbs et al., 1974), and other morphological traits (Hubbs and Miller, 1948; Woodman, 1992; Scott and Crossman, 1998). Such high levels of differentiation in genetic and morphological traits suggest that multiple cryptic species exist within the current concept of *Rhinichthys osculus* (Page and Burr, 1998; Billman et al., 2010).

Here, we study the phylogenetic relationships and genetic diversity of speckled dace throughout Oregon's Great Basin. During the Pleistocene pluvial period (beginning 2.5 million years ago), this region contained many large lakes which experienced periods of repeated connection and isolation that ended about 10,000 years ago (Hubbs and Miller, 1948). The complex geologic history of this area has created complicated and under-characterized patterns of speciation and species boundaries (Smith, 1978; Johnson, 2002; Smith et al., 2002). We aim to determine whether Pleistocene basin boundaries structured the genetic diversity of speckled dace in Oregon's Great Basin, and ultimately, which groups of populations warrant taxonomic recognition as full species.

1.2. Foskett Spring speckled dace

The genetic relationships and taxonomic status of dace in Oregon's Great Basin hold substantial conservation relevance because of the presence of Foskett Spring speckled dace (*Rhinichthys osculus* ssp.), a putative subspecies that has never been formally described. This population was listed as threatened under the federal Endangered Species Act in 1985 because of its extremely isolated range, low numbers of individuals, and vulnerable spring habitat (USFWS, 1985, 2009). Foskett Spring speckled dace is represented by a single population from Foskett Spring (located on the west side of Coleman Lake in Lake County, Oregon). Coleman Valley is a small endorheic basin within Warner Basin (Fig. 1) that Hubbs and Miller (1948, p. 65) suggested was last flooded by an arm of Warner Lake at the end of the pluvial period (9000–10,000 years ago). It is likely that speckled dace became isolated in Foskett Spring at that time (Ardren et al., 2010).

Our current understanding of the taxonomic status, morphology, and ecology of Foskett Spring speckled dace is limited. The only known morphological description is a personal communication from Oregon State University (OSU) professor Carl Bond, dated 1990 (USFWS, 1998). This indicated that Foskett Spring speckled dace differed slightly from those found in the rest of Warner Basin by possessing a shorter lateral line, larger eye, and a more posterior dorsal fin origin. Recently, Ardren et al. (2010) examined two mitochondrial genes (Cytb and ND2) for speckled dace throughout the Warner and Goose Lake drainages and found that within Warner Basin, Foskett Spring dace were the most divergent population, but did not show reciprocal monophyly (usually a requirement for species or subspecies status when using genetic data; Zink, 2004). Ardren et al. (2010) concluded it was "difficult to justify subspecies status" for Foskett dace using their data. The lack of reciprocal monophyly suggests incomplete lineage sorting of the ancestral genotypes and supports their suggestion that isolation events within Warner Basin were recent. However, in order to determine whether speckled dace from Foskett Spring are genetically diagnosable and what taxonomic status or protection this entity warrants, additional genetic data and analyses, including non-mitochondrial loci, wider geographic sampling, and explicit evaluation of gene-flow among populations are required.

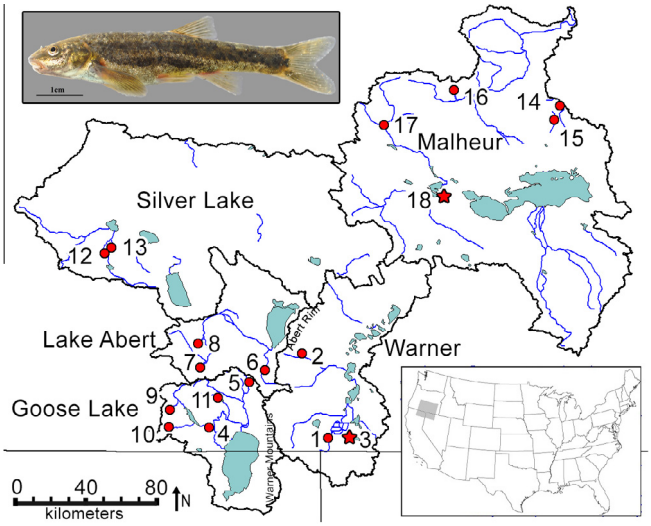

Fig. 1. A map of the study area showing the Eastern Oregon endorheic basins investigated in this study (outlined in bold black lines). Sampling locations are indicated by dots (the two springs are starred): 1 = Twentymile (Warner), 2 = Snyder (Warner), 3 = Foskett Spring (Warner), 4 = Drew's Creek (Goose Lake), 5 = Cox Creek (Goose Lake), 6 = Crooked Creek (Lake Abert), 7 = Little Coffee Pot Creek (Lake Abert), 8 = South Creek (Lake Abert), 9 = Cottonwood Creek (Goose Lake), 10 = Dent Creek (Goose Lake), 11 = Hay Creek (Goose Lake), 12 = Silver Creek (Silver Lake), 13 = West Fork Silver (Silver Lake), 14 = Cow Creek (Malheur), 15 = Rattlesnake (Malheur), 16 = Sawtooth Creek (Malheur), 17 = Nicoll Creek (Malheur), 18 = Stinking Lake Spring (Malheur). Bottom inset shows location of sample area within the United States. Top inset: picture of a speckled dace from Goose Lake basin.

1.3. Research questions

Here we use the phylogenetics and population genetics of speckled dace in Oregon's Great Basin region to answer three major questions: (1) What is the pattern of genetic differentiation among populations of speckled dace and how does it relate to the geologic history of the paleodrainages? (2) Which units within the present concept of *R. osculus* warrant recognition as species? (3) Should Foskett Spring speckled dace be recognized as a distinct taxonomic entity? This study will aid a future taxonomic revision of speckled dace, inform management decisions, and help reveal the biogeographic history of the region.

2. Materials and methods

2.1. Sampling locations

Sampling sites were located in Warner Valley and all surrounding basins containing speckled dace: Goose Lake, Lake Abert, Silver Lake, and Malheur basins (Fig. 1). Speckled dace are not known to occur in Catlow and Long Valleys, which also border the Warner Valley. We sampled up to 15 individuals from at least three streams in each basin, as well as 41 Foskett Spring speckled dace (Fig. 1, Table A.1). We also sampled spring dace from Stinking Lake Spring in Malheur drainage, which was the only other known spring population of speckled dace in the basins of interest (based on museum records and input from district biologists).

2.2. Specimen acquisition

Specimens were captured using a seine net, dipnet, or minnow trap, and euthanized in an aqueous solution of tricaine methane

sulfonate (MS-222, approximately 2 g/L), buffered with equal concentration of sodium bicarbonate. Following euthanization, small pieces of tissue (fin clips or small plugs of epaxial muscle weighing 1 g or less) were taken from each individual and preserved in 95% ethanol or saturated salt (SED) buffer. These collections were performed with approval from the Oregon State University Institutional Animal Care and Use Committee (ACUP #4050) and under permit from the Oregon Department of Fish and Wildlife (STP 15423 and 16430).

2.3. DNA extraction, sequencing, and genotyping

DNA was extracted using an NaCl extraction protocol adapted from Lopera-Barrero et al. (2008). We amplified the mitochondrial NADH subunit 2 (ND2, 1001 bp) and the more slowly evolving (Brown et al., 1979) nuclear S7 intron (857 bp). Ardren et al. (2010) also employed ND2 in their study of speckled dace from this region and found it to contain substantial phylogenetic signal. S7 is commonly used in ichthyological studies (Wang et al., 2002; Luca et al., 2008; Houston et al., 2010) and pilot sequencing indicated it would show enough genetic divergence to be informative.

ND2 was amplified using the primers ASN (5′-CGCGTTTAG CTGTTAACTAA-3′) and ILE (5′-CCGGATCACTTTGATAGAGT-3′) (Dr. Thomas Dowling, Arizona State University, pers. comm.). The PCR reactions contained 1× PCR buffer, 1.5 mM $MgCl_2$, 1 μM of each primer, 0.2 mM dNTPs, and 0.5 U PlatinumTaq (Invitrogen). The temperature profile consisted of an initial denaturation at 95 °C for 5 min; 35 cycles of 95 °C for 30 s, annealing at 60 °C for 1 min, and extension at 72 °C for 1.5 min; followed by a final extension at 72 °C for 10 min. S7 was amplified using S7RPEX1F (5′-TGGCCTCTTCCTTGGCCGTC-3′) and S7RPEX2R (5′-AACTGTCTG GCTTTTCGCC-3′) (Chow and Hazama, 1998). The PCR reactions contained 1× Qiagen MasterMix and 0.5 μM of each primer. The temperature protocol was as follows: 95 °C for 5 min; 35 cycles of 95 °C for 30 s, annealing between 59 and 62 °C for 1 min, and extension at 72 °C for 2 min; followed by a final extension at 72 °C for 10 min.

Following amplification, all PCR products were purified using ExoSAP (Hanke and Wink, 1994) and then sequenced in both directions using BigDye Terminator v3.1 Cycle Sequencing (ABI) on the ABI 3730 in the CoreLabs at Oregon State University. Forward and reverse sequences were edited in Geneious v6.1.5 (created by Biomatters); heterozygotes in the nuclear gene were labeled with IUPAC ambiguity codes. The sequences for ND2 and S7 were deposited in GenBank under numbers KF421693–KF421812 and KF421595–KF421692, respectively, and alignments for each gene were deposited in Dryad (http://datadryad.org; doi: http://dx.doi.org/10.5061/dryad.v64q8). The number of haplotypes per population for ND2 and S7 is presented in Table A.3.

Eight microsatellite loci were genotyped for 223 individuals using the primers listed in Table A.2. PCRs were performed with 1× GoTaq Master Mix (Promega) and 0.3–0.5 μM primer concentration (forward primer fluorescently labeled). The temperature protocol was as follows: 95 °C for 5 min; 35 cycles of 95 °C for 30 s, annealing temp for 1 min (see Table A.2), and extension at 72 °C for 2 min; followed by a final extension at 72 °C for 10 min. These PCR products were genotyped on an ABI 3730xl (Hatfield Marine Science Center, Oregon State University). GeneMapper v4.1 software (Life Technologies) was used to assign allele peaks and assignments were checked by eye. The number of alleles for the 8 microsatellite loci by population and allelic richness by rarefaction is presented in Table A.3.

All loci were tested for possible null alleles and allelic dropout using MicroChecker 2.2.4 (van Oosterhout et al., 2004). We used GENEPOP v3.4 (Raymond and Rousset, 1995) to test for linkage disequilibrium and deviations from Hardy–Weinberg equilibrium (HWE) amongst loci, with Fisher's exact tests of 1000 dememorization steps, 100 batches, and 1000 iterations per batch for all combinations of loci.

2.4. Phylogenetic analysis

Sequences of *Rhinichthys atratulus* (blacknose dace) and *R. cataractae* (longnose dace) were used as outgroups for all phylogenetic analyses. The *R. cataractae* individuals were collected from Big Wood River in Idaho (provided by Tom Dowling, Arizona State University). *R. atratulus* sequences were downloaded from GenBank (accession numbers: GU134262.1, AY103163.1, JN569221.1, JN569262.1).

Partition schemes for the nuclear and mtDNA alignments were selected through PartitionFinder v1.0.1 (Lanfear et al., 2012). jModelTest (Posada, 2008) was used to select the models of nucleotide substitution for the alignments using Bayesian information criterion (BIC), corrected Akaike information criterion (AICc) and decision theory (DT) methods (after the likelihood scores were calculated using a BIONJ base tree). For the nuclear (S7) dataset, all three methods chose F81+G (all nucleotide substitution rates are equal, base frequencies are allowed to vary), and for the mitochondrial (ND2) dataset all three methods chose TrN+G.

The Maximum Likelihood (ML) phylogeny was inferred in RAxML v7.2.6 (Stamatakis, 2006) using separate partitions for each ND2 codon position, and a single partition for the S7 intron, as determined by PartitionFinder. A rapid bootstrap analysis and search for the best-scoring ML tree was performed with 1000 bootstrap replicates, the GTR+G model of rate heterogeneity, 25 distinct rate categories, random seed bootstrapping, and random number seed for MP starting tree computations. We analyzed ND2 and S7 separately and in concatenation. The resulting trees were viewed in FigTree v1.3.1 (http://tree.bio.ed.ac.uk/software/figtree/).

The Bayesian phylogenetic analyses were performed in MrBayes v3.2.1 (Ronquist et al., 2012), with partitioning as in the ML analysis and using the evolutionary models selected by jModelTest. Because MrBayes does not implement TrN+G (which is a special case of GTR+G), GTR+G (nst = 6, rates = gamma) was used for the ND2 dataset, and F81+I+G (nst = 1, rates = invgamma) was used for S7. The MCMCMC algorithm employed three heated chains and one cold chain, and 1 million generations sampled every 500 generations. To assess the convergence of model parameters and tree topology after the initial 1 million runs, we examined whether the standard deviation of split frequencies between the two runs was less than 0.01, and added 500,000 additional generations until the standard deviation was below 0.01 and stationarity was achieved. Twenty-five percent of the generations were discarded as the burn-in. Results from all phylogenetic analyses were viewed in FigTree v1.3.1.

We used the Bayesian BEAST algorithm (Heled and Drummond, 2010) to date clade divergences and coalescence times to the most recent common ancestor for different clades. BEAUti 1.7 (Drummond and Rambaut, 2007) was used to generate the input file for BEAST 1.6.1 (Drummond and Rambaut, 2007). The evolutionary model was set to TrN+G for ND2 and F81+G for S7, as selected by jModelTest (above). The clock model was set to lognormal relaxed clock (uncorrelated) based upon a lognormal prior using the 'speciation birth death' process. Ucld.mean was a diffuse gamma distribution (shape 0.001, scale 1000), initial value set centrally within the prior distribution. BEAST runs consisted of 2.0×10^8 MCMC generations with sampling every 10,000 generations. The output log file from each run was examined in Tracer (Rambaut and Drummond, 2007) and the posterior distributions were checked for stationarity; where effective sample sizes less than 200 for all parameters of interest. A maximum clade

credibility tree (1% burn-in) was summarized in TreeAnnotator v1.5.4. The resulting trees were viewed in FigTree v1.3.1.

Our BEAST tree was calibrated at the split between *Rhinichthys cataractae* and *R. osculus*, with a prior uniform distribution mean of 6.3 mya, a maximum of 8.1 mya, and a minimum >4.5 mya. The oldest fossils assignable to *Rhinichthys osculus* were identified by Smith et al. (2002), and came from the Glenns Ferry Formations on the Snake River Plain (about 4.5 mya). Smith and Dowling (2008) calculated the estimated time of origin for the *R. osculus* clade using Marshall (2008)'s method for bracketing divergence times, where a confidence interval is calculated using multiple fossils from multiple dates. There were only three time horizons containing *R. osculus*, so the 50% alpha value was quite large (1.8 my, Smith and Dowling, 2008), and resulted in an estimate of 6.3 mya for the divergence of *R. osculus* from its sister taxon, *R. cataractae*. The fossil provides a strict minimum age for the group that contains this fossil (in this case *Rhinichthys osculus*). We used age priors with uniform distributions in which the minimum limit is set to the minimum age of the stage where the fossil is found (4.5 mya) and the upper limit is set high enough to not restrict the upper bound (8.1 mya, 2 C.I.s above the minimum).

2.5. Population level analysis

To investigate population level genetic patterns, the number of parsimony informative sites (ND2 and S7) was calculated in MEGA 5.05 (Tamura et al., 2011). Nucleotide and haplotype diversity were calculated in DNAsp v5 (Rozas, 2009). Genetic distances based on a Neighbor-Joining tree (Tamura-Nei) were calculated between and within clades (determined by the ML and Bayesian analyses) in MEGA 5.05 (Tamura et al., 2011) using a *p*-distance model. A Mantel test was performed in R (R Core Team, 2012) to test for isolation by distance using the genetic distance (from MEGA) and geographical distances (strait lines between sample sites). Mantel's test is a randomization test that calculates the probability that two distance matrices are more similar than expected by chance alone. Statistical significance was tested using 10,000 random permutations.

Using the genetic distances above, a haplotype network of the ND2 haplotypes was created for the three Warner Valley locations using statistical parsimony in TCS v1.21 (Clement et al., 2000), which provided a closer look at structuring of haplotypes within that basin. A S7 haplotype network included all samples from Goose Lake, Silver Lake, Lake Abert, and Warner Valley to see if there is structuring within this group.

Population level genetic structure was also investigated using microsatellite genotypes. The number of populations (K) that best explained the pattern of genetic variability was estimated using the program Structure v2.3.3 (Pritchard et al., 2000) under an admixture model with correlated allele frequencies and a burn-in length of 500,000 and 5,000,000 MCMC steps. The average admixture coefficient for individuals (alpha) was inferred with a uniform prior for alpha (initial value = 1, max = 10, SD = 0.025). We tested $K = 1–20$ with 4 runs and saw that there was no population signal above $K = 10$. So with putative Ks (2–12), 20 independent runs were performed. The second order rate of change of the likelihood function with respect to K (ΔK) was plotted to aid in the selection of the optimal number of populations (following Evanno et al., 2005).

Genetic differentiation between pairs of populations (each basin and each spring, as identified by Structure) was estimated with FST (Weir and Cockerham, 1984) in GENEPOP v3.4, and Nei's D genetic distance (Nei et al., 1983) was computed with the GENDIST package in Phylip (Felsenstein, 1993). Observed (H_O) and expected (H_E) heterozygosities were calculated in Arlequin 3.5.1.3 (Excoffier et al., 2005; Table A.3). A Factorial Correspondence Analysis (FCA) was performed in GENETIX 4.03 (Belkhir et al., 1996).

3. Results

3.1. Phylogenetics and major divergences

Maximum likelihood and Bayesian phylogenetic analyses of S7 (Fig. 2) reveal the presence of three clades within Oregon's Great Basin: speckled dace from Malheur basin, those from Stinking Lake Spring, and those from the other four basins combined (Goose Lake, Silver Lake, Lake Abert, and Warner, inclusive of Foskett Spring). All three clades are reciprocally monophyletic (with posterior probability values greater than 0.95), with Malheur and Stinking Lake Spring inferred as sister clades, although with low support for this relationship (bootstrap of 74, posterior probability of 0.80). The average percent sequence divergences among these three clades shows low levels of intra-basin differentiation that are nevertheless significantly higher than within-basin differentiation (2-sample *t*-test, $p < 0.0001$). Dace from Stinking Lake Spring average 1.56% and 1.40% difference from the four basins and from Malheur, respectively, and Malheur fish differ on average by 1.17% from the other four basins.

Factorial Correspondence Analysis of the microsatellite data indicates the same three distinct and completely non-overlapping clusters; dace from Malheur, Stinking Lake Spring, and the other basins combined (Fig. 3). Stinking Lake Spring dace separate from the cluster of Goose Lake, Silver, Warner, and Lake Abert along the first FCA axis, while Malheur dace are separate from all other dace along the second and third FCA axis.

Bayesian and Maximum likelihood analyses of ND2 (Fig. 4) also recognize the same three major clades, but further subdivide the last of these into daces from Goose Lake, those from Silver Lake, and a clade consisting of those from Lake Abert and Warner Valley (including Foskett Spring), with Lake Abert forming its own clade within a paraphyletic Warner group. The deepest divergence splits the Malheur clade from all other speckled daces in the region of interest. When branches with less than 75% bootstrap support are collapsed, Goose Lake, Silver Lake, and Lake Abert/Warner, form a polytomy of unknown relationships. There is no further structuring within each basin based on the ND2 phylogeny, as the individual stream populations do not form clusters. Average percent sequence divergences (Table 1) reveals the greatest genetic differentiation is among Malheur, Stinking Lake Spring, and the clade containing the other four basins (which corresponds to the three major groups identified by the S7 gene tree and the FCA). Sequence divergences among the Goose Lake, Silver Lake, and Lake Abert/Warner clades range from 4.23% to 5.72%. The higher percent divergences observed from ND2 versus S7 are expected given the higher rate of evolution of mitochondrial loci when compared to nuclear loci (Brown et al., 1979) and is also reflected in the number of parsimony informative sites (292 out of 1001 bp for ND2, 73 out of 857 bp for S7).

The BEAST analysis (Fig. 5) has the same tree topology as the ML and MrBayes analyses of the concatenated dataset, and reveals estimated divergences between clades of speckled dace first occurring in the Pliocene, with subsequent divergences occurring in the Pleistocene (Fig. 5). Malheur stream fish are estimated to have diverged from the clade containing all other speckled dace examined in this study during the Pliocene (between 1.91 and 5.38 mya), and Stinking Lake Spring dace most likely diverged from the other basins at the Pliocene–Pleistocene boundary (between 1.31 and 3.97 mya). Further diversification occurred during the Pleistocene, with the Warner/Lake Abert clade splitting from the Silver Lake/Goose Lake clade between 0.71 and 2.33 mya. Lake

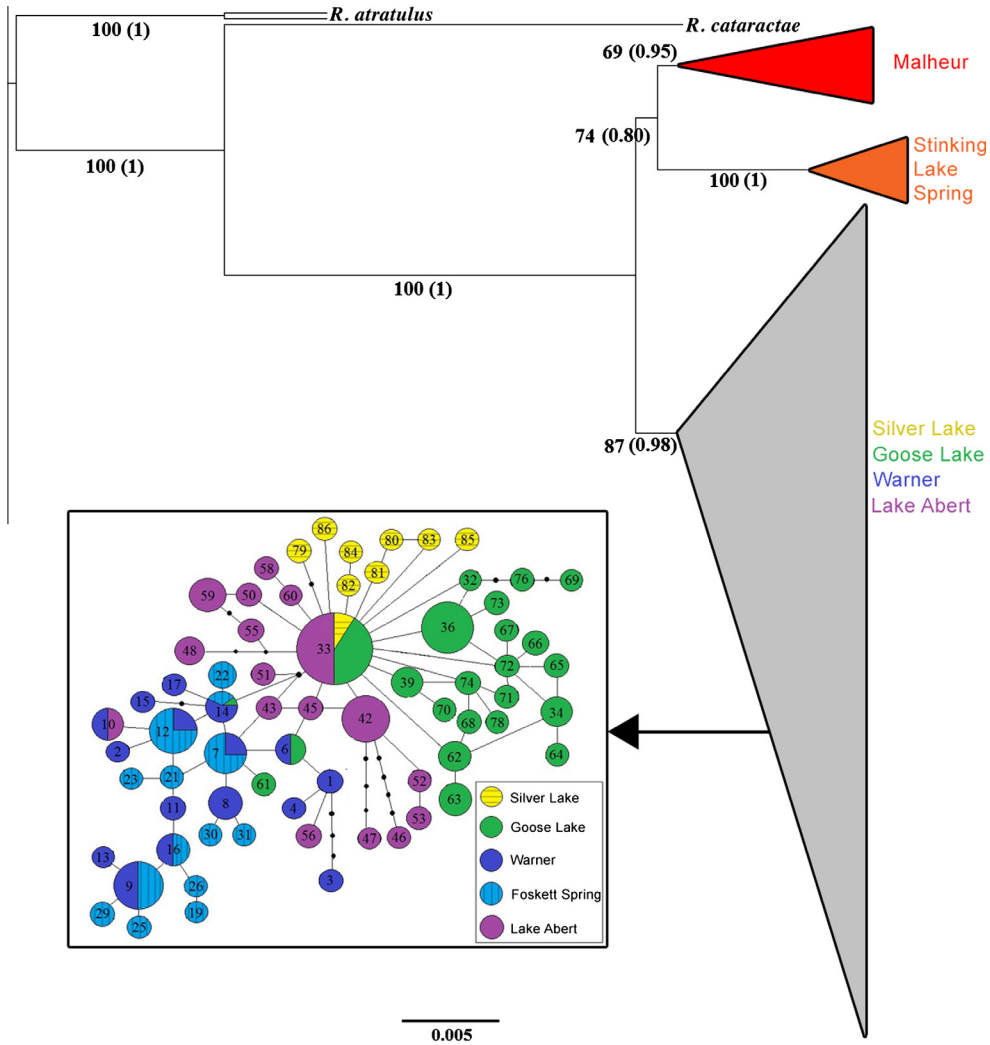

Fig. 2. Maximum likelihood (RAxML) phylogeny based on S7 sequences from 161 individuals of *Rhinichthys osculus* collected from Oregon's Great Basin. As the Bayesian phylogenetic reconstruction showed the same topology, only the ML tree is presented here. Bootstrap values and Bayesian posterior probabilities from MrBayes in parentheses are presented at branches. The scale bar shows the number of substitutions per sequence position. Inset: Haplotype network of Silver Lake, Goose Lake, Warner, Foskett Spring, and Lake Albert individuals based on S7 sequences (large clade). Circles are shaded by the location of specimens, and black dots show inferred substitutions. Each line-segment represents a single base pair change out of the 857 base pairs sequenced. Numbers in the circles indicate the arbitrary haplotype number which matches the GenBank definition. The relative size of the circles indicates the number of individuals with a given haplotype.

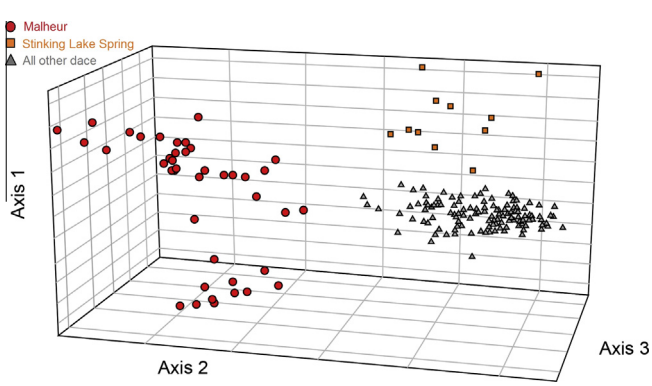

Fig. 3. Factorial correspondence analysis plot based on microsatellite data of the *Rhinichthys osculus* collected from Southeast Oregon. Circles represent Malheur, squares represent Stinking Lake Spring, and triangles represent the other four basins combined (Goose Lake, Silver Lake, Warner, and Lake Abert). Scaling is not uniform among axes.

Abert split from Warner around 0.23 mya and Goose Lake and Silver Lake diverged around 1.09 mya.

3.2. Population level differentiation

The parsimony haplotype network of S7 haplotypes from the Goose Lake/Silver Lake/Lake Abert/Warner clade (Fig. 2 inset) shows some further genetic structuring among these basins. The most common haplotype is shared by Goose Lake, Silver Lake, and Lake Abert individuals, but these three basins do not share any other haplotypes and each basin shows slight clustering. Warner Valley individuals share some haplotypes with Goose Lake; however, the Warner haplotypes mostly appear nested within the Lake Abert haplotypes. The scarcity of shared haplotypes indicates that these basins experience restricted gene-flow, despite not having had sufficient evolutionary time to achieve reciprocal monophyly at the S7 locus. The ND2 haplotype network of Warner Valley individuals (Fig. 4 inset) shows no further structure in this

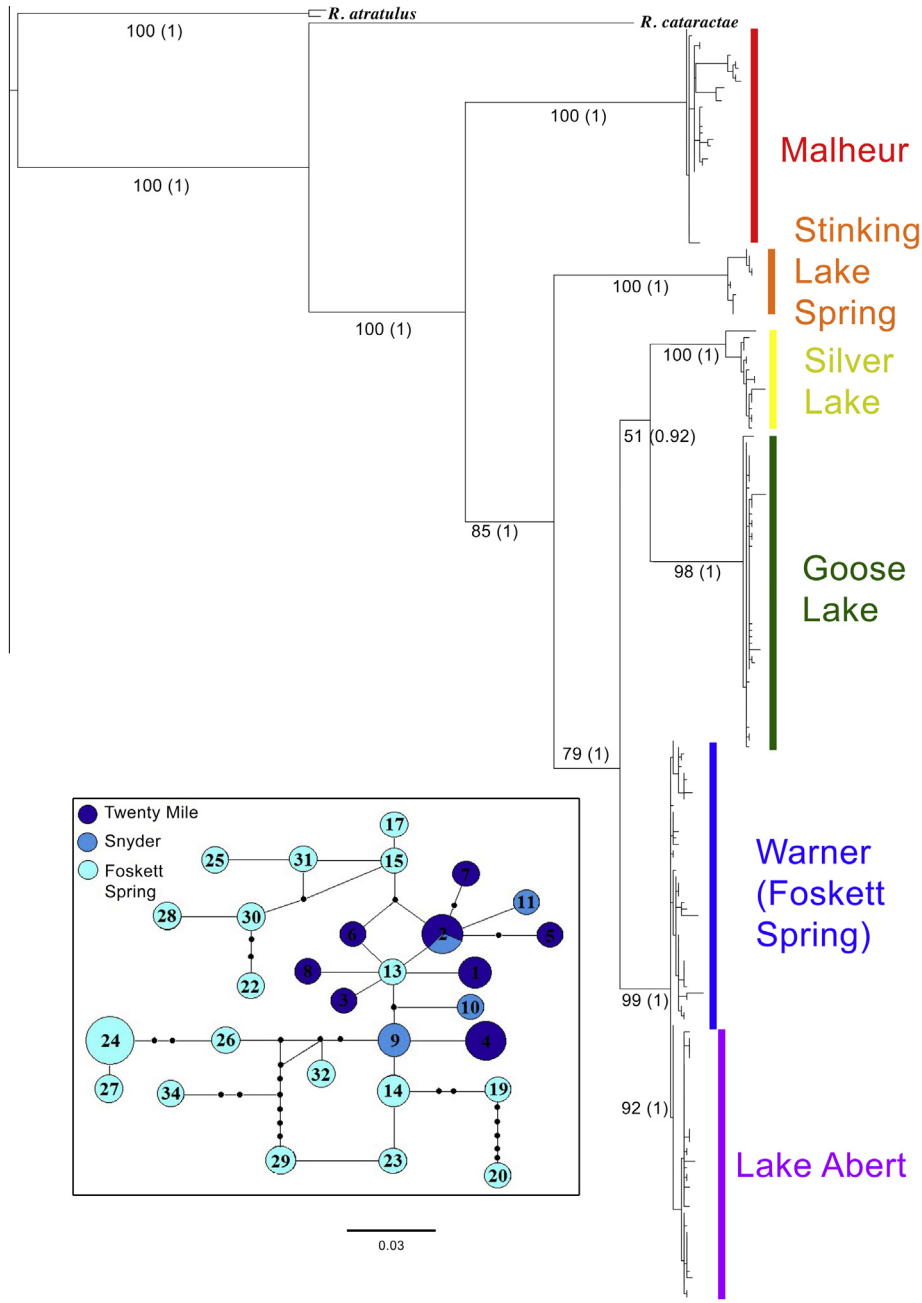

Fig. 4. Maximum likelihood (RAxML) phylogeny based on the ND2 sequences from 197 individuals of *Rhinichthys osculus* collected from Eastern Oregon's Great Basin. As the Bayesian phylogenetic reconstruction showed the same topology, only the ML tree is presented here. Bootstrap values and Bayesian posterior probabilities from MrBayes in parentheses are presented at branches. The scale bar shows the number of substitutions per sequence position. Inset: Haplotype network of Warner Valley individuals based on ND2 sequences. Circles are colored by the location of specimens, and black dots show inferred substitutions. Numbers in the circles indicate the arbitrary haplotype number which matches the GenBank definition. The relative size of the dots indicates the number of individuals with a given haplotype. A line between circles represents a single base pair change out of the 1001 base pairs sequenced. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 1
Percent sequence divergence of ND2 based on average number of different base pairs per 1000 total base pairs. Percentages in italics along the diagonal represent intra-basin variation.

	Warner	Lake Abert	Goose Lake	Silver Lake	Stinking Lake Spring	Malheur
Warner	*1.67*	1.89	4.23	5.72	10.00	12.99
Lake Abert		*1.28*	4.27	5.68	9.95	12.94
Goose Lake			*2.53*	4.96	9.97	12.96
Silver Lake				*0.81*	12.84	13.12
Stinking Lake					*0.45*	10.83
Malheur						*4.75*

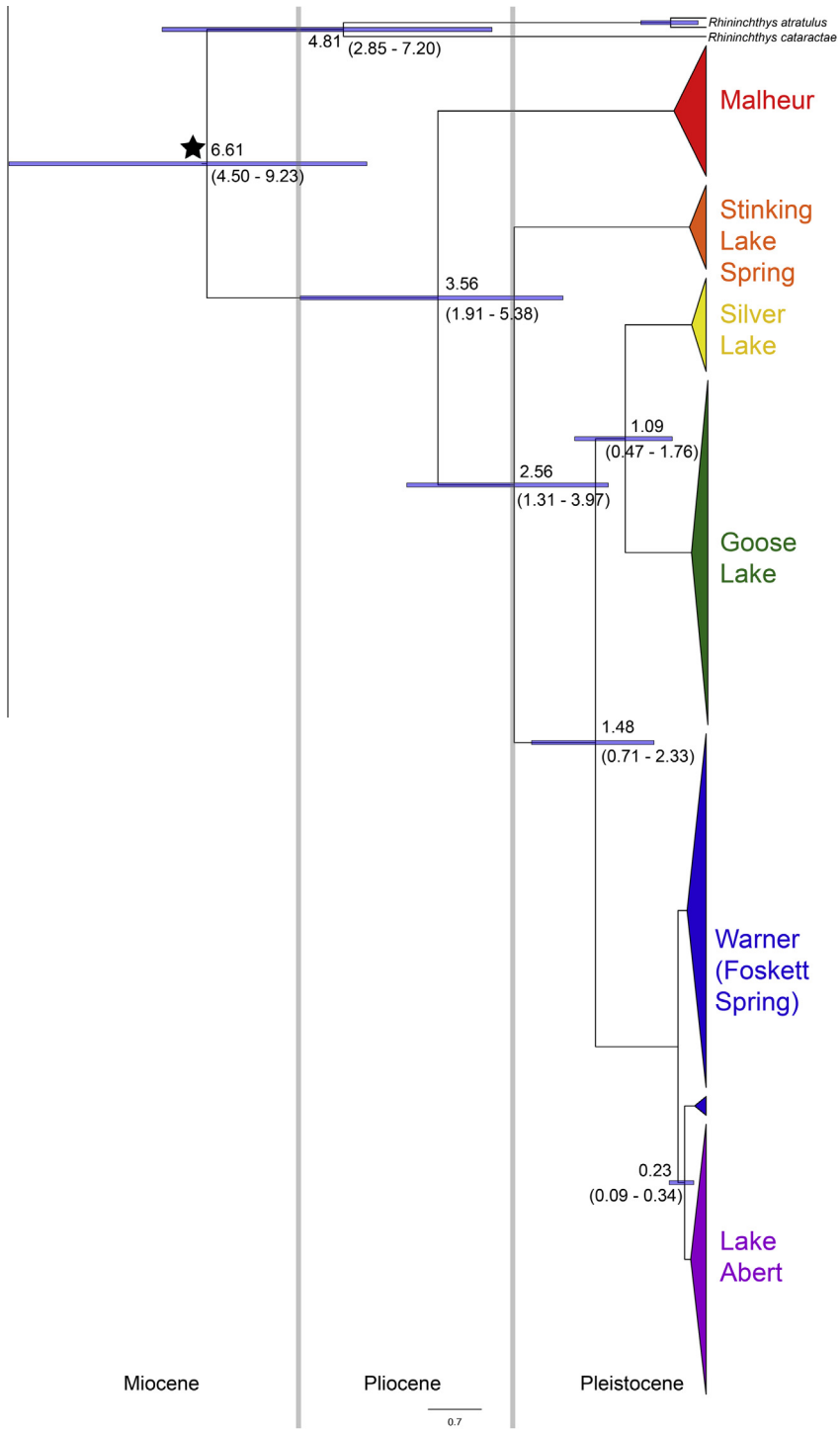

Fig. 5. BEAST tree using the concatenated dataset (ND2 and S7) of *Rhinichthys osculus* collected from Southeast Oregon. The dates at branching points are in millions of years before present; bars indicate 95% HPD of divergence dates. Geologic epochs are separated by vertical bars. The fossil calibrated node is indicated with a star.

basin. Haplotypes from Foskett Spring speckled dace are scattered among others from the two Warner Valley streams. However, the network also highlights the fact that Foskett Spring speckled dace share no ND2 haplotypes with dace from other Warner locations.

Population-level genetic structuring, the finest-scale in this study, is revealed by the Structure analysis of microsatellites. Using the Evanno et al. (2005) method, Structure infers seven populations in our sample area; Malheur, Stinking Lake Spring, Silver Lake, Goose Lake, Warner, Foskett Spring, and Lake Abert (Fig. 6). In contrast to our other analyses, Structure inferred Foskett Spring speckled dace as a distinct population, implying no recent gene-flow between that spring and the surrounding Warner Basin population. The isolation by distance test (Mantel test) revealed that genetic distance is significantly associated with geographic distance ($r = 0.840$; $p < 0.001$), indicating that these seven populations of speckled dace are geographically structured.

Pairwise F_{ST} with corresponding significance and Nei's D are shown in Table 2. The highest pairwise F_{ST}s among populations are found between Malheur and Foskett, Lake Abert, and Stinking Lake; between Stinking Lake and Foskett Spring and Lake Abert;

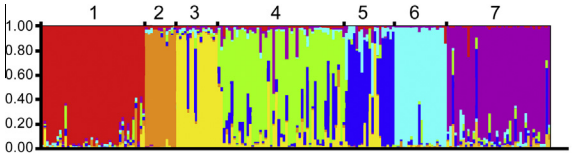

Fig. 6. Structure bar plot results obtained on the microsatellite dataset of *Rhinichthys osculus* collected from Southeast Oregon at $K = 7$, 1 is Malheur, 2 is Stinking Lake Spring, 3 is Silver Lake, 4 is Goose Lake, 5 is Warner, 6 is Foskett Spring, and 7 is Lake Abert. Each vertical bar represents one individual and the color of the bar represents the population assignment. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

and between Foskett Spring and Lake Abert. Nei's D shows similar patterns; high differentiation among Malheur and Stinking Lake, Foskett, and Lake Abert, and very low differentiation between Goose Lake and Silver Lake. Although geographically separate, speckled dace from Silver Lake and Goose Lake are more similar than speckled dace from the neighboring Foskett Spring, Warner, and Lake Abert are to each other.

4. Discussion

This study represented the first phylogeographic investigation of speckled dace (*Rhinichthys osculus*) in Oregon's Great Basin. We identified three distinctive evolutionary lineages (Malheur, Stinking Lake, and a lineage spanning the four other basins) that diverged during the Pliocene or early Pleistocene and should likely be elevated to species-status once their full geographic extent is discovered, and their morphological diversity described. Many of the observed genetic divergences are consistent with the complex geographic history of the region, but unexpectedly, one of the earliest diverging lineages is limited geographically to Stinking Lake Spring, which is of late Pleistocene origin and much younger than the age of the relict lineage that it now harbors. In agreement with Ardren et al. (2010) we determined that Foskett Spring speckled dace is not monophyletic, and does not warrant species or subspecies status. However, due to its lack of gene-flow with other populations, numerous unique haplotypes, unusual habitat and geographically circumscribed range, we recommend that Foskett Spring speckled dace be recognized as an Evolutionarily Significant Unit within the broader concept of *Rhinichthys osculus*. These results and recommendations are discussed further below.

4.1. Phylogeography in Oregon's Great Basin

The initial diversification of speckled dace in Oregon's Great Basin likely occurred before the start of the Pleistocene and resulted in three main clades that are currently distributed in streams within the Malheur drainage, Stinking Lake Spring, and the other four basins (Fig. 5). The Malheur stream dace may have separated from all others as early as the Miocene–Pliocene boundary (Fig. 5). This early separation makes sense geologically, because the Malheur drainage was not connected to these other Great Basin drainages during the Pluvial period (Hubbs and Miller, 1948), and was more likely recently connected to the Snake drainage immediately to its eastern border.

Further diversification within the westernmost basins occurred during the Pleistocene, with the dace in the Goose Lake and Silver Lake drainages splitting from those in Warner and Lake Abert around 1.5 mya. This date is unexpectedly recent, given that the Warner Mountain uplift (which now separates the Goose and Warner drainages) occurred around 4.5 mya (Minckley et al., 1986). That barrier would have been significant enough to prevent genetic mixing between Goose Lake and Warner Valley, but our analysis of sequence data infers that it took until 2.33 mya at the earliest for the genetic split between Goose Lake and Warner to occur.

The BEAST analysis infers a split of the Lake Abert basin dace from the Warner Valley dace around 230,000 years ago, more than a million years after the split of these two basins from the Goose Lake/Silver Lake clade. The close affinity between Lake Abert and Warner daces is intriguing, because Lake Abert basin is located between Silver Lake and Goose Lake (Fig. 1) and is separated by low elevation barriers (250 m to Silver Lake and 300 m to Goose Lake) relative to Abert Rim (760 m) which separates Lake Abert from the Warner Valley. Despite its height, the Abert Rim fault uplifted around only 175,000 years ago (Pezzopane and Weldon, 1993). That date corresponds closely with the inferred date of genetic division of Lake Abert from the Warner Valley.

Although it is clear that three highly divergent evolutionary lineages of speckled dace exist within Oregon's Great Basin, the geographic limits of each lineage are currently unclear, and they may be closely related to dace in adjacent drainages due to ancient watershed connections. For example, Silver Lake drainage is thought to have been connected in the north to the Deschutes and Columbia systems, or with weaker support, to the Klamath system (Hubbs and Miller, 1948). Goose Lake likely drained into the Pit River system to the south before becoming isolated (Hubbs and Miller, 1948). This connection is supported by evidence from a study of tui chub (*Gila bicolor*), a species which also has a widespread distribution throughout the Great Basin. Tui chub from Goose Lake and Pit River were found to be indistinguishable from each other based on microsatellites (Chen et al., 2009). Cytochrome b sequences for *Rhinichthys osculus* reported by Ardren et al. (2010) suggest that Goose Lake dace are closely related to one of the linages found in Klamath basin, and although they did not sample Silver Lake or the Pit River system, their data provide some support for a close linkage among the Klamath, Goose Lake, and Silver Lake basins. These biogeographic patterns suggest that dace in surrounding drainages, such as the Klamath, Deschutes, Snake, or Lahontan, may be the closest relatives of the speckled dace in Oregon's Great Basin, but only a widespread, range-wide phylogeographic study will fully resolve these systematic questions.

Table 2
F_{ST} estimates between populations (below the diagonal) and Nei's D (above the diagonal). Values with an asterisk indicate table-wide statistical significance of $p < 0.05$ after the Bonferroni correction for multiple comparison.

	Malheur	Stinking Lake Spring	Silver Lake	Goose Lake	Warner	Foskett Spring	Lake Abert
Malheur		0.543	0.448	0.343	0.478	0.574	0.540
Stinking Lake	0.109*		0.421	0.473	0.454	0.531	0.462
Silver Lake	0.088*	0.081*		0.262	0.401	0.501	0.310
Goose Lake	0.069	0.086*	0.004		0.360	0.442	0.281
Warner	0.097*	0.090*	0.076	0.067		0.300	0.318
Foskett Spring	0.114*	0.109*	0.097*	0.083*	0.060		0.497
Lake Abert	0.117*	0.101*	0.066	0.061	0.071*	0.180*	

4.2. Unrecognized taxonomic entities

Three lineages of speckled dace were consistently identified and diagnosibly different despite the genetic marker and analysis method used. Malheur stream dace, Stinking Lake Spring dace, and dace from the other four endorheic basins collectively are reciprocally monophyletic in phylogenies derived from nuclear DNA and mtDNA, thus fulfilling a major requirement for species status using the phylogenetic species concept (deQueiroz and Donoghue, 1990). The percent sequence divergence among these clades is high; Malheur basin dace are about 12% different from other clades on average (ND2), while Stinking Lake Spring dace are about 10% different at that locus. By comparison, for ND2, European *Barbus* species differ by an average of 7.97% (Zardoya and Doadrio, 1998) and species in the North American cyprinid genus *Lythrurus* differ by an average of 10.3% (Schmidt et al., 1998). The level of sequence divergence for S7 that we observed between the three main clades (1.17–1.56%) is close to the levels observed between other sister species within the family Cyprinidae (Wang et al., 2002). Since the mitochondrial and nuclear data indicate that the divergence between the Malheur clade, Stinking Lake Spring clade, and the four-basin clade are on the order of species-level differences noted in other studies of cyprinids, we recommend species status for these entities once their geographic limits inside and outside of Oregon are fully documented.

The taxonomic history of speckled dace is littered with available names for species and subspecies that are currently in the synonymy of *Rhinichthys osculus*, some of which may need to be resurrected and applied to the lineages revealed by this study. However, only a detailed morphological and geographic comparison of historic type specimens with a broad sample of recently-collected material will be able to sort out which of these names are valid. Until that comparison can be completed, we refrain from any formal changes to taxonomy.

Each of the Goose Lake, Silver Lake, and Lake Abert/Warner basin daces are monophyletic based on mtDNA and all four basins were identified as separate populations by the microsatellite-based Structure analysis. However, the levels of divergence between these basins are lower than typical between-species differences and the nuclear data do not distinguish them, so we consider these to all be distinct populations of the same species. These populations may be considered for Evolutionarily Significant Unit designation if they are biologically significant (Haig et al., 2006), because of their potential to become a species given sufficient time (Darwin's incipient species; Avise and Hamrick, 1996). However, not enough is currently known about the ecology, morphology, or biological significance of the speckled dace in these basins to determine whether ESU status is warranted.

4.2.1. Stinking Lake Spring speckled dace

The unexpected discovery of a 2.5 million year old lineage of speckled dace isolated in the 10,000 year old Stinking Lake Spring merits discussion and explanation. Stinking Lake Spring dace are genetically much more distinct from Malheur stream dace than Foskett Spring dace are from the rest of Warner, despite the similar geologic history of these two basins. Both Malheur and Warner basin were filled with large Pleistocene lakes which have been desiccating since approximately 10,000 years ago (Hubbs and Miller, 1948). Therefore, the dace populations in Foskett Spring and Stinking Lake Spring have likely been isolated in their current locations for a similar amount of time and it follows that the genetic divergence of Stinking Lake Spring Dace is too extreme to have been caused by this relatively short period of isolation.

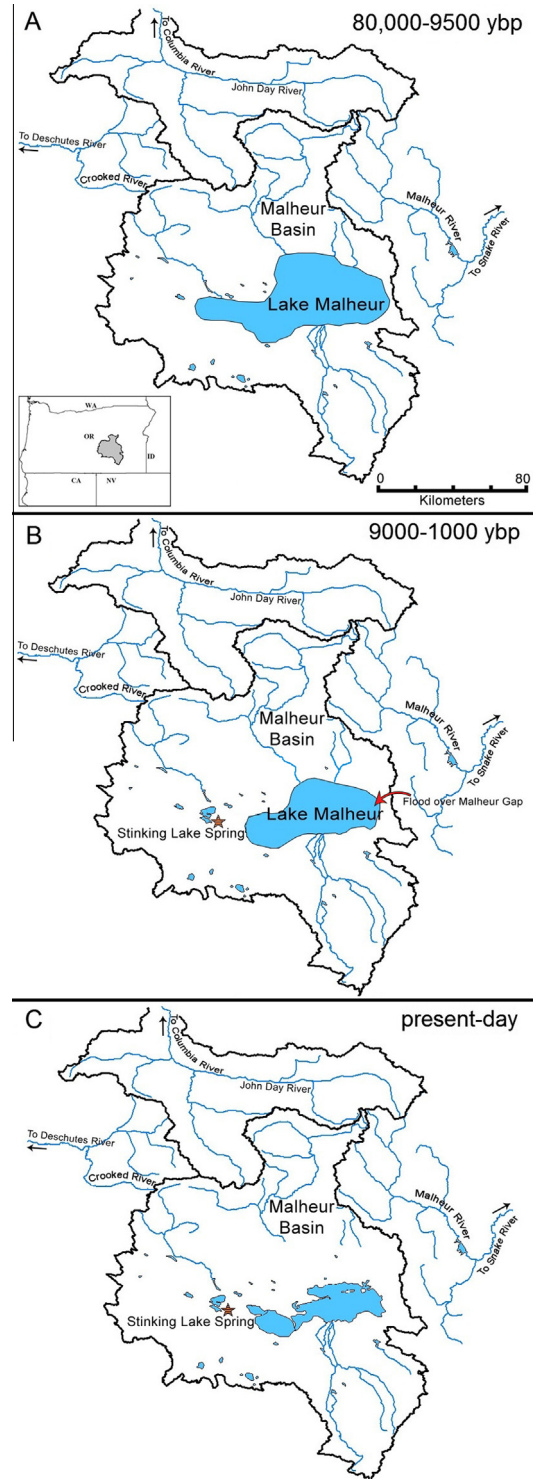

Fig. 7. Hydrological history of the Malheur basin and neighboring portions of the Malheur, John Day, and Crooked River drainages over the last 10,000 years. The star indicates Stinking Lake Spring. Inset shows location of Malheur Basin in Eastern Oregon. (Panel A) Historic high water levels of Lake Malheur in Malheur basin (9500 years before present) and historic river connections. We hypothesize that one lineage of speckled dace occurred throughout the whole basin at this time. (Panel B) Malheur Basin during the period of desiccation which began about 9000 years ago. Spillover from the Malheur River across the Malheur Gap may have occurred during this period of drying, leading to the introduction of a new lineage of speckled dace into Lake Malheur. (Panel C) Present-day water levels showing isolation of the rivers and the springs. Two divergent lineages of speckled dace now occur in the Basin, one in the northern streams and one in the isolated springs.

We consider population replacement to be the most likely explanation for the current isolation of the highly distinctive Stinking Lake Spring dace, given the tumultuous past of the Great Basin region. In this scenario, the Stinking Lake Spring dace are relicts of a once widespread population of speckled dace that occurred throughout Malheur when the basin was filled with a large lake (Fig. 7A). As the lake desiccated (about 9000 years ago, Fig. 7B), individuals in the rivers (in the northern Ochoco and southern Steens ranges) would have become isolated from the dace in Stinking Lake Spring. During high water events or floods, speckled dace could have invaded the streams in the basin from the nearby Malheur River in the Snake River Basin by way of the low elevation Malheur Gap (Dugas, 1998) or from the John Day system by way of stream capture. This invading population would have spread to occupy the streams in the basin, but would not have been able to reach the now-isolated springs. This scenario would result in two distinct lineages that have evolved separately since the separation of Malheur Basin and Snake River drainages, but currently occur in close proximity (Fig. 7C). It would predict that stream dace in Malheur would be more closely related to the dace from the Snake drainage than to Stinking Lake Spring speckled dace. A future phase of this study will sample dace from all basins bordering Malheur to test these predictions and determine if a population invasion has occurred.

A similar explanation involves population introgression, where an invasion of speckled dace from outside of Malheur basin could have resulted in mitochondrial genome replacement through numerical bias of the invaders, as occurred in *Cyprinodon* in Mexico (Carson and Dowling, 2006). However, the mtDNA, nuclear sequence, and microsatellite loci show the same pattern of high divergence between Stinking Lake Spring and Malheur stream dace, so there is no evidence supporting a case of population introgression and mitochondrial replacement.

A third, but less likely explanation involves rate heterogeneity. It is possible that the spring habitat elevates rates of molecular evolution due to low N_e or increased mutation rates, either of which would cause more rapid genetic divergence. Small springs will usually have smaller population sizes than large streams, and small effective population sizes have been associated with accelerated rates of DNA sequence evolution (Ohta, 1992; Johnson and Seger, 2001; Woolfit and Bromham, 2003). Additionally, spring environments often have high temperatures because of warm water aquifers or lack of water flow. High temperatures will result in increased metabolic rates among poikilotherms, which may result in elevated rates of mutation stemming from the by-products of oxidative respiration (Rand, 1994; Martin, 1999; Gillooly et al., 2005). Though initially appealing, an explanation based on rate heterogeneity seems unlikely because Foskett Spring has likely maintained similar population sizes to Stinking Lake Spring, but shows much lower levels of genetic divergence. Similarly, Foskett Spring is known to have constant, warm temperatures (18 °C, pers. comm. Paul Scheerer, ODFW), but does not appear to demonstrate a particularly fast rate of genetic divergence.

Finally, the deeper genetic divergence found between Stinking Lake Spring and Malheur dace than between Foskett Spring and Warner dace might also be a result of increased selection pressure. If Stinking Lake Spring and Malheur streams present very different selective environments, it could cause genes under selection to diverge more rapidly. ND2 (NADH subunit 2) is a functional, protein-coding gene that may be affected by different selection pressures. However, S7 is a non-coding nuclear intron and is thought to be neutral (Chow and Hazama, 1998), so should not be affected by selection. This neutrally evolving marker still shows the pattern of deep divergence between Stinking Lake Spring dace and the surrounding Malheur populations, implying that differential selection pressures is not the main cause of the unique evolutionary pattern found in Stinking Lake Spring speckled dace.

4.3. Foskett Spring

We found no genetic evidence that speckled dace from Foskett Spring warrant subspecies or species status. Foskett Spring speckled dace are not monophyletic, but rather appear within a larger clade also including specimens from the rest of Warner Basin (Fig. 4). The lack of substantial genetic differentiation between Foskett Spring and Warner basin dace based on both nuclear and mtDNA sequence data indicates that this population has not been isolated long enough to diverge significantly. However, we did find evidence that Foskett Spring dace are a distinct population separate from the rest of Warner Valley dace, having no shared mitochondrial haplotypes from 41 individuals (Fig. 4 inset) and being separated by the Structure analysis of the microsatellite data (Fig. 6). The genetic isolation and the unique habitat of the population qualify Foskett Spring dace for Evolutionarily Significant Unit (ESU) status. Foskett Spring dace will likely have continued Federal protection as a threatened entity due to the extremely restricted and isolated range, and low population numbers.

5. Conclusion and future directions

We found clear evidence for the presence of three distinct taxonomic entities of Pliocene age within the present concept of *Rhinichthys osculus* in Oregon's Great Basin: dace from Malheur, dace from Stinking Lake Spring, and dace from the other four endorheic basins combined (Goose Lake, Silver Lake, Lake Abert, and Warner, inclusive of Foskett Spring). These taxonomic units may include dace in surrounding basins, therefore formal designations for these species will require expanded sampling to determine their geographic limits, as well as careful morphological comparison with the type material for the many species and subspecies currently in the synonymy of *Rhinichthys osculus*. The

Table A.1
Basin and stream locations (springs marked by asterisks) of the speckled dace collected for this study, with number of vouchers taken at each location. Samples are deposited in the Oregon State Ichthyology Collection under the catalog numbers provided. Full locality data is available at http://www.fishnet2.net/search.aspx?c=os.

Basin	Location	# tissues	Catalog number
Warner Valley	Twentymile Creek	15	OS 18125
	Snyder Creek	6	OS 18126
	*Foskett Spring	41	OS 18127
	(total)	62	
Goose Lake	Cox Creek	3	OS 18129
	Drew's Creek	15	OS 18128
	Cottonwood Creek	11	OS 18400
	Dent Creek	14	OS 18393
	Hay Creek	15	OS 18394
	(total)	58	
Lake Abert	Crooked Creek	15	OS 18130
	Little Coffeepot Creek	15	OS 18391
	South Creek	15	OS 18397
	(total)	45	
Silver Lake	Silver Creek	8	OS 18401
	West Fork Silver	8	OS 18396
	(total)	16	
Malheur	Cow Creek	11	OS 18398
	Rattlesnake Creek	4	OS 18389
	Sawtooth Creek	15	OS 18390
	Nicoll Creek	10	OS 18392
	*Stinking Lake Spring	12	OS 18286
	(total)	52	

microsatellite data revealed further population structure, and distinguished each sampled basin and spring. Dace from Goose Lake, Silver Lake, Lake Abert, Warner Valley, and Foskett Spring have not had enough evolutionary time to diverge along the slower evolving nuclear intron marker, but are clearly distinct populations. The mitochondrial and nuclear sequence data indicate that Foskett Spring speckled dace are not a distinct species, but the evidence of recent genetic isolation and the unique habitat of the population qualify it for designation as an evolutionarily significant unit.

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Appendix A

Tables A.1–A.3.

Appendix B. Supplementary material

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.ympev.2014.04.027.

Table A.2
Primer sequence, annealing temperatures (−0.5 indicates a touch-down protocol of 0.5 °C per cycle), number of alleles per locus, allele size ranges, fluorescent dye used, and original references for the eight microsatellite primers used in this study.

Locus	Primer sequences	Annealing temp.	# Of alleles	Size range	Dye	Source
CypG3	F: AGTAGGTTTCCCAGCATCATTGT R: GACTGGACGCCTCTACTTTCATA	67 (−0.5)	65	126–436	6-FAM	Baerwald and May (2004)
CypG9	F: GCAGTCACGTATTAAGGCGAGCAG R: GAGCGGACTCTCAGGCACCTACC	67 (−0.5)	6	95–115	VIC	Baerwald and May (2004)
CypG13	F: ACACCCAGTTCCTGATGGA R:CATTTATTTCATACTGCACTACACA	55	60	164–456	VIC	Baerwald and May (2004)
CypG24	F: CTGCCGCATCAGAGATAAACACTT R: TGGCGGTAAGGGTAGACCAC	67 (−0.5)	24	150–230	PET	Baerwald and May (2004)
CypG27	F: AAGGTATTCTCCAGCATTTAT R: GAGCCACCTGGAGACATTACT	57	26	214–318	NED	Baerwald and May (2004)
CypG33	F: TATGAGCTTTGGAAAGAGACACTG R:AATAGCCGGGAAATTATCAATAGA	67 (−0.5)	3	84–92	NED	Baerwald and May (2004)
Lco1	F: CACGGGACAATTTGGATGTTTTAT R: GGGGGCAGCATACAAGAGACAAC	59	77	277–597	PET	Turner et al. (2004)
Lco4	F: ATCAGGTCAGGGGTGTCACG R: TGTTTATTTGGGGTCTGTGT	58	28	228–288	6-FAM	Turner et al. (2004)

Table A.3
Number of haplotypes per population for ND2 and S7 with number of private haplotypes in parentheses, and the nucleotide diversity in each population. Number of alleles for the 8 microsatellite loci by population (Allelic richness by rarefaction in parenthesis) and the observed and expected heterozygosities by population based on the microsatellite data.

Gene (length)	Total # of haplotypes	Malheur	Stinking Lake Spring	Silver Lake	Goose Lake	Lake Abert	Warner	Foskett Spring
ND2 (1001 bp)	120	25 (25)	6 (6)	11 (11)	25 (25)	17 (17)	12 (12)	23 (23)
Nucleotide diversity	0.06066	0.00500	0.00560	0.00097	0.00385	0.00112	0.00160	0.00474
Haplotype diversity	0.973	0.811	0.891	0.958	0.774	0.889	0.837	0.932
S7 (857 bp)	98	7 (7)	5 (5)	9 (8)	28 (20)	21 (15)	16 (9)	18 (9)
Nucleotide diversity	0.00532	0.00293	0.00063	0.00264	0.00088	0.00126	0.00144	0.00124
Haplotype diversity	0.758	0.675	0.473	0.964	0.544	0.526	0.729	0.738
Microsatellite (repeat motif)	Total # of alleles							
CypG3 (TAGA)	65	21 (13.4)	10 (11.3)	15 (13.0)	25 (13.6)	21 (11.3)	14 (7.4)	10 (11.6)
CypG9 (CAGA)	6	5 (3.0)	3 (4.4)	5 (3.0)	3 (2.0)	4 (3.0)	2 (3.3)	3 (4.0)
CypG13 (TAGA)	60	32 (15.3)	11 (10.5)	23 (18.0)	45 (17.8)	33 (15.2)	21 (11.5)	15 (15.7)
CypG24 (CAGA)	24	10 (6.7)	8 (8.0)	12 (10.6)	15 (8.5)	7 (10.3)	13 (6.6)	9 (5.4)
CypG27 (TAGA)	26	9 (13.5)	14 (13.2)	17 (13.9)	23 (13.5)	21 (13.8)	18 (9.8)	12 (13.4)
CypG33 (CTGT)	3	2 (1.9)	1 (1.0)	1 (1.0)	2 (2.0)	2 (1.9)	2 (2.0)	2 (1.3)
Lco1 (GATA)	77	25 (14.1)	17 (17.0)	18 (15.1)	39 (17.4)	37 (16.1)	20 (14.3)	20 (17.4)
Lco4 (GTGA)	26	6 (4.4)	4 (4.0)	7 (6.6)	23 (12.4)	13 (7.9)	9 (5.7)	6 (7.3)
H_O (S.D.)		0.7205 (0.2717)	0.8496 (0.1621)	0.9280 (0.0667)	0.7331 (0.2801)	0.7314 (0.3141)	0.7374 (0.2770)	0.7866 (0.3141)
H_E (S.D.)		0.7611 (0.2671)	0.8447 (0.0975)	0.8847 (0.1166)	0.8004 (0.2768)	0.7413 (0.3155)	0.7608 (0.3034)	0.7520 (0.1994)

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